How Often Are Two Random Permutations Comparable?

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joint work with Jonathan Boretsky (McGill University), Alvaro Cornejo (University of Kentucky), Paul Horn (University of Denver), Nathan Lesnevich (Oklahoma State University), and Tyrrell McAllister (University of Wyoming)

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Counting intervals in a poset

For a finite poset P with partial order $\leq_P$, define its set of intervals by

$$\mathrm{Int}(P) = \{(x, y) \in P^2 : x \leq_P y\}.$$

Thus

$$|\mathrm{Int}(P)| = \sum_{y \in P} |\{x \in P : x \leq_P y\}|.$$

This is a natural statistic for any family of posets

$$P_1, P_2, P_3, \ldots$$

and one can ask for exact formulas or the asymptotics of

$$|\mathrm{Int}(P_n)|.$$

Examples: from Boolean lattices to Tamari lattices

Boolean lattice

Let B_n be the poset of all subsets of $[n]$, ordered by inclusion.
Its intervals are pairs

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since each element of $[n]$ is in neither set, in B only, or in both A and B.

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Tamari lattice

The Tamari lattice is a Catalan poset: its elements may be viewed as triangulations of a polygon, or equivalently as binary trees.
Chapoton proved the interval formula

$$|\operatorname{Int}(\mathrm{Tam}_n)| = \frac{2(4n+1)!}{(n+1)!(3n+2)!}.$$

The symmetric group and its natural orders

Write a permutation $w \in \mathfrak{S}_n$ in one-line notation:

$$w = w_1 w_2 \cdots w_n.$$

The inversion number is

$$\ell(w) = |\{(i,j) : i < j \text{ and } w_i > w_j\}|.$$

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Strong Bruhat order

Strong Bruhat order is generated by covers

$$w \lessdot_B wt_{ij}, \qquad 1 \leq i < j \leq n, \qquad \ell(wt_{ij}) = \ell(w) + 1,$$

where wt_{ij} is obtained from w by swapping the entries in positions i and j. Geometrically, this is the closure order on Schubert cells in the complete flag variety.

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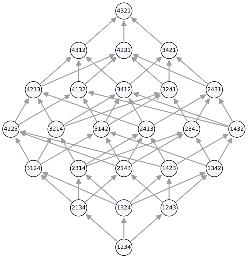
In particular,

$$u \preceq_W w \qquad \Longrightarrow \qquad u \le_B w.$$

A first comparison: $\mathfrak{S}_4$



Strong Bruhat order on $\mathfrak{S}_4$

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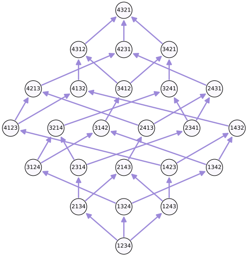
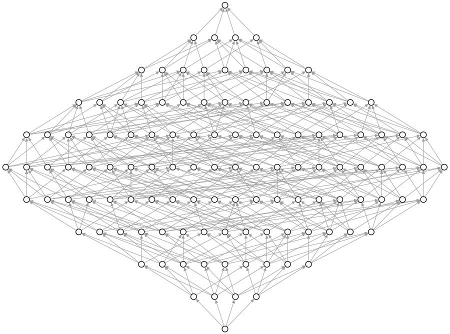


Right weak order on $\mathfrak{S}_4$

Size comparison

	cover relations	$\lvert \mathrm{Int}(\cdot) \rvert$
strong Bruhat order	58	213
right weak order	36	151

The same comparison for $\mathfrak{S}_5$



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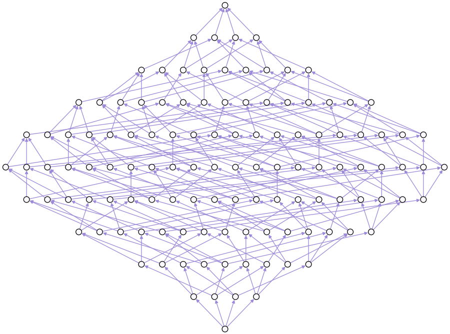
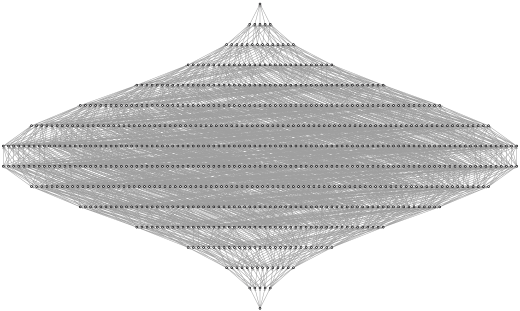


Right weak order on $\mathfrak{S}_5$

Size comparison

	cover relations	$\lvert\mathrm{Int}(\cdot)\rvert$
strong Bruhat order	444	3781
right weak order	240	1899

And for $\mathfrak{S}_6$



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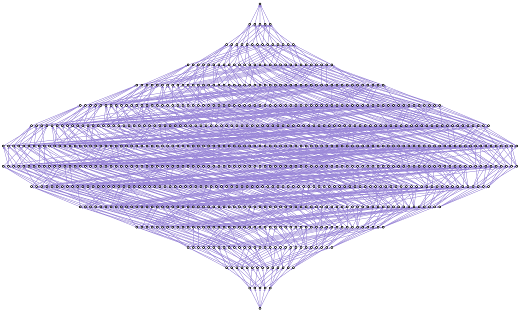

Right weak order on $\mathfrak{S}_6$

Size comparison

	cover relations	$\lvert\mathrm{Int}(\cdot)\rvert$
strong Bruhat order	3708	98407
right weak order	1800	31711

Excuse me, I was promised a talk on probability

Counting intervals is the same as comparability probability

Let P be a finite poset. If $x, y \in P$ are chosen independently and uniformly, then

$$\mathbb{P}(x \leq_P y) = \frac{|\operatorname{Int}(P)|}{|P|^2}.$$

Thus, once the size of the poset is known, counting intervals is equivalent to computing the probability that two random elements are comparable.

Main result

Reframing interval counting probabilistically, we ask for the chance that two independent random permutations are comparable in weak order.

Theorem (Boretsky–Cornejo–H–Horn–Lesnevich–McAllister, *2026*)

$$\mathbb{P}(\sigma_1 \preceq_W \sigma_2) = \frac{|\operatorname{Int}(\mathfrak{S}_n, \preceq_W)|}{(n!)^2} = \exp\left(\left(-\frac{1}{2} + o(1)\right) n \log n\right).$$

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Equivalently,

$$|\operatorname{Int}(\mathfrak{S}_n, \preceq_W)| = (n!)^2 \exp\left(\left(-\tfrac{1}{2} + o(1)\right) n \log n\right).$$

Hammett–Pittel previously proved

$$\exp((-1 + o(1)) n \log n) \leq \mathbb{P}(\sigma_1 \preceq_W \sigma_2) \leq \exp(-\Theta(n)).$$

How should one read this asymptotic?

On the $n \log n$ scale,

$$(n!)^2 = \exp\Big((2 + o(1)) n \log n\Big).$$

Our theorem gives

$$|\operatorname{Int}(\mathfrak{S}_n, \preceq_W)| = \exp\Big((\tfrac{3}{2} + o(1)) n \log n\Big).$$

Hammett–Pittel's lower bound gave

$$|\operatorname{Int}(\mathfrak{S}_n, \preceq_W)| \geq \exp\Big((1 + o(1)) n \log n\Big).$$

Since Hammett–Pittel's upper bound was $\exp(-\Theta(n))$ for the probability, after multiplying by $(n!)^2$ it has the same leading $n \log n$-scale exponent as the trivial upper bound $(n!)^2$.

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Scale picture

$$\begin{aligned}
\underbrace{\exp\Big((1 + o(1)) n \log n\Big)}_{\text{HP lower}} \leq |\operatorname{Int}(\mathfrak{S}_n, \preceq_W)| &= \underbrace{\exp\Big((\tfrac{3}{2} + o(1)) n \log n\Big)}_{\text{BCHHLM}} \\
&\leq \underbrace{\exp\Big((2 + o(1)) n \log n\Big)}_{\text{HP upper}} = \underbrace{\exp\Big((2 + o(1)) n \log n\Big)}_{\text{all } (n!)^2 \text{ ordered pairs}}.
\end{aligned}$$

Weak order and permutation posets

The key step is to replace weak-order intervals by linear extensions of a poset attached to a permutation.

Permutation poset

For $\sigma \in \mathfrak{S}_n$, define P_σ on ground set $[n]$ by

$$i \prec_\sigma j \qquad \Longleftrightarrow \qquad i < j \text{ and } \sigma(i) < \sigma(j).$$

These dimension-two posets are often called permutation posets; they appear naturally in Björner–Wachs' work on permutation statistics and linear extensions.

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Write $e(Q)$ for the number of linear extensions of a finite poset Q.

Key reduction

For every $\omega \in \mathfrak{S}_n$,

$$|[\mathrm{id}, \omega]_{\preceq_W}| = e(\mathsf{P}_{\omega^{-1}}).$$

Therefore

$$|\operatorname{Int}(\mathfrak{S}_n, \preceq_W)| = \sum_{\omega \in \mathfrak{S}_n} |[\mathrm{id}, \omega]_{\preceq_W}| = \sum_{\omega \in \mathfrak{S}_n} e(\mathsf{P}_{\omega^{-1}}) = \sum_{\sigma \in \mathfrak{S}_n} e(\mathsf{P}_\sigma).$$

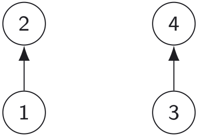
Example: the interval below 3412

Take

$$\tau = 3412, \qquad \tau^{-1} = 3412.$$

Since $\tau = \tau^{-1}$, the relevant permutation poset is simply P_{3412}.
The permutation poset P_{3412} on $[4]$ has relations

$$1 \prec_{3412} 2, \qquad 3 \prec_{3412} 4.$$



A linear extension is any total order with 1 before 2 and 3 before 4. Hence the linear extensions are

$$1234, \quad 1324, \quad 1342, \quad 3124, \quad 3142, \quad 3412.$$

These are exactly the permutations π such that $\pi \preceq_W 3412$.

Takeaway

The lower interval $[\mathrm{id}, 3412]_{\preceq_W}$ is encoded by the linear extensions of P_{3412}.

Linear extensions: the new problem

The reduction turns the problem into estimating

$$e(\mathrm{P}_\sigma)$$

for a typical permutation $\sigma \in \mathfrak{S}_n$.

Greene–Kleitman–Fomin data

For a finite poset Q on n elements, define

$$A_k(Q) = \max\{|A_1 \cup \cdots \cup A_k| : A_i \text{ antichains}\},$$

and

$$C_k(Q) = \max\{|C_1 \cup \cdots \cup C_k| : C_i \text{ chains}\}.$$

With $A_0(Q) = C_0(Q) = 0$, define partitions by first differences:

$$a_k(Q) = A_k(Q) - A_{k-1}(Q), \qquad c_k(Q) = C_k(Q) - C_{k-1}(Q).$$

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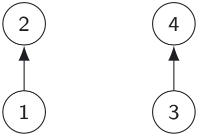
Theorem (Bochkov–Petrov)

$$\prod_i a_i(Q)! \le e(Q) \le \frac{n!}{\prod_j c_j(Q)!}.$$

Example: GKF data and Bochkov–Petrov for P_{3412}

For P_{3412} we have two disjoint chains, so the structure is very easy to read off.



Antichains and chains

The largest antichain has size 2 (for instance $\{1, 3\}$), and two antichains cover all four elements. Thus

$$A_1 = 2, \quad A_2 = 4, \qquad \text{so} \qquad (a_1, a_2) = (2, 2).$$

Likewise the largest chain has size 2, and two chains cover all four elements. Thus

$$C_1 = 2, \quad C_2 = 4, \qquad \text{so} \qquad (c_1, c_2) = (2, 2).$$

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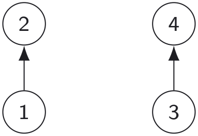

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Bochkov–Petrov

Hence

$$2!\,2! \leq e(\mathsf{P}_{3412}) \leq \frac{4!}{2!\,2!},$$

so

$$4 \leq e(\mathsf{P}_{3412}) \leq 6.$$

In fact, from the previous slide we know $e(\mathsf{P}_{3412}) = 6$.

GKF data for the permutation poset

For permutation posets, the Greene–Kleitman–Fomin data become very concrete.

Fact 1: the lower-bound data

A chain in P_σ is exactly an increasing subsequence of σ. Hence

$$\mathrm{ht}(\mathrm{P}_\sigma) = \mathrm{LIS}(\sigma).$$

Thus by Mirsky's theorem, if $\mathrm{LIS}(\sigma) \le t$, then P_σ is a union of t antichains. Equivalently, the antichain partition satisfies

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Fact 2: the upper-bound data

Let

$$\lambda(\sigma) = (\lambda_1, \lambda_2, \ldots)$$

be the RSK shape of σ. Greene's theorem gives

$$C_k(\mathrm{P}_\sigma) = \lambda_1 + \cdots + \lambda_k,$$

so $c_k(\mathrm{P}_\sigma) = \lambda_k$.
Therefore Bochkov–Petrov gives

$$e(\mathrm{P}_\sigma) \le \frac{n!}{\prod_k \lambda_k!}.$$

Switching to probability

The interval count and comparability probability are related by

$$\mathbb{P}(\sigma_1 \preceq_W \sigma_2) = \frac{|\operatorname{Int}(\mathfrak{S}_n, \preceq_W)|}{(n!)^2}.$$

Using the reduction to permutation posets,

$$|\operatorname{Int}(\mathfrak{S}_n, \preceq_W)| = \sum_{\sigma \in \mathfrak{S}_n} e(\mathsf{P}_\sigma).$$

Therefore

$$\mathbb{P}(\sigma_1 \preceq_W \sigma_2) = \frac{1}{(n!)^2} \sum_{\sigma \in \mathfrak{S}_n} e(\mathsf{P}_\sigma).$$

What Bochkov–Petrov gives

For every $\sigma \in \mathfrak{S}_n$,

$$\prod_i a_i(\mathsf{P}_\sigma)! \leq e(\mathsf{P}_\sigma) \leq \frac{n!}{\prod_k c_k(\mathsf{P}_\sigma)!}.$$

Lower bound

Lower bound: typical longest increasing subsequence

Baik–Deift–Johansson implies that for a uniform permutation,

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Then

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Now fix $\sigma \in T$. By Fact 1, P_σ has height at most t, so Mirsky's theorem partitions P_σ into t antichains.
Writing their sizes as

$$a_1 + \cdots + a_t = n,$$

Bochkov–Petrov gives

$$e(\mathrm{P}_\sigma) \geq \prod_{i=1}^{t} a_i!.$$

Lower bound: balancing the antichains

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$$n = qt + r, \qquad 0 \le r < t.$$

Among all nonnegative integer t-tuples with sum n, the product $\prod_i a_i!$ is minimized when the entries are as equal as possible. Hence

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Therefore

$$\begin{aligned}
\mathbb{P}(\sigma_1 \preceq_W \sigma_2) &= \frac{1}{(n!)^2} \sum_{\sigma \in \mathfrak{S}_n} e(\mathrm{P}_\sigma) \\
&\ge \frac{1}{(n!)^2} \sum_{\sigma \in T} \prod_{i=1}^{t} a_i! \\
&\ge \frac{|T|}{(n!)^2} (q!)^{\,t-r}((q+1)!)^{\,r} = (1 - o(1)) \frac{(q!)^{\,t-r}((q+1)!)^{\,r}}{n!}.
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\end{aligned}$$

Since $t \sim \sqrt{n}$ and $q \sim \sqrt{n}$, Stirling yields

$$\log \mathbb{P}(\sigma_1 \preceq_W \sigma_2) \ge \left(-\tfrac{1}{2} + o(1)\right) n \log n.$$

Upper bound via RSK

RSK and Plancherel measure

The Robinson–Schensted correspondence gives a bijection

$$\mathfrak{S}_n \longleftrightarrow \bigsqcup_{\lambda \vdash n} \mathrm{SYT}(\lambda) \times \mathrm{SYT}(\lambda).$$

Thus each permutation σ has an associated partition

$$\lambda(\sigma) = (\lambda_1, \lambda_2, \ldots),$$

called its RSK shape.

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Plancherel measure

For a uniformly chosen permutation $\sigma \in \mathfrak{S}_n$ and a partition $\lambda \vdash n$,

$$\mathbb{P}(\lambda(\sigma) = \lambda) = \frac{(f^\lambda)^2}{n!}, \qquad f^\lambda = |\mathrm{SYT}(\lambda)|.$$

By the hook-length formula,

$$f^\lambda = \frac{n!}{\mathrm{hookprod}(\lambda)}, \qquad \text{so} \qquad \mathbb{P}(\lambda(\sigma) = \lambda) = \frac{n!}{\mathrm{hookprod}(\lambda)^2}.$$

Here hookprod(λ) denotes the product of the hook lengths of λ.

Upper bound via RSK shape

Conditioning by RSK shape, using that RSK shape is invariant under inverses, and applying Fact 2 together with Bochkov–Petrov, we get

$$\mathbb{P}(\sigma_1 \preceq_W \sigma_2 \mid \lambda(\sigma_2) = \lambda) = \frac{\frac{1}{(n!)^2} \sum_{\substack{\omega \in \mathfrak{S}_n \\ \lambda(\omega)=\lambda}} |[\mathrm{id}, \omega]_{\preceq_W}|}{\frac{|\{\omega \in \mathfrak{S}_n : \lambda(\omega) = \lambda\}|}{n!}}$$

$$= \frac{1}{|\{\omega \in \mathfrak{S}_n : \lambda(\omega) = \lambda\}|} \sum_{\substack{\omega \in \mathfrak{S}_n \\ \lambda(\omega)=\lambda}} \frac{|[\mathrm{id}, \omega]_{\preceq_W}|}{n!}$$

$$= \frac{1}{|\{\omega \in \mathfrak{S}_n : \lambda(\omega) = \lambda\}|} \sum_{\substack{\omega \in \mathfrak{S}_n \\ \lambda(\omega)=\lambda}} \frac{e(\mathsf{P}_{\omega^{-1}})}{n!} \leq \frac{1}{\prod_{k \geq 1} \lambda_k!}.$$

Upper bound via RSK shape

Conditioning by RSK shape, using that RSK shape is invariant under inverses, and applying Fact 2 together with Bochkov–Petrov, we get

$$\mathbb{P}(\sigma_1 \preceq_W \sigma_2 \mid \lambda(\sigma_2) = \lambda) = \frac{\dfrac{1}{(n!)^2} \sum_{\substack{\omega \in \mathfrak{S}_n \\ \lambda(\omega)=\lambda}} |[\mathrm{id}, \omega]_{\preceq_W}|}{\dfrac{|\{\omega \in \mathfrak{S}_n : \lambda(\omega) = \lambda\}|}{n!}}$$

$$= \frac{1}{|\{\omega \in \mathfrak{S}_n : \lambda(\omega) = \lambda\}|} \sum_{\substack{\omega \in \mathfrak{S}_n \\ \lambda(\omega)=\lambda}} \frac{|[\mathrm{id}, \omega]_{\preceq_W}|}{n!}$$

$$= \frac{1}{|\{\omega \in \mathfrak{S}_n : \lambda(\omega) = \lambda\}|} \sum_{\substack{\omega \in \mathfrak{S}_n \\ \lambda(\omega)=\lambda}} \frac{e(\mathrm{P}_{\omega^{-1}})}{n!} \leq \frac{1}{\prod_{k \geq 1} \lambda_k!}.$$

Group by RSK shape

$$\mathbb{P}(\sigma_1 \preceq_W \sigma_2) = \sum_{\lambda \vdash n} \mathbb{P}(\lambda(\sigma_2) = \lambda)\, \mathbb{P}(\sigma_1 \preceq_W \sigma_2 \mid \lambda(\sigma_2) = \lambda)$$

$$\leq \sum_{\lambda \vdash n} \frac{n!}{\mathrm{hookprod}(\lambda)^2} \frac{1}{\prod_{k \geq 1} \lambda_k!}.$$

Upper bound: reduction to a deterministic partition estimate

Rewrite the previous bound as

$$\mathbb{P}(\sigma_1 \preceq_W \sigma_2) \le n! \sum_{\lambda \vdash n} e^{-\Psi(\lambda)},$$

where

$$\Psi(\lambda) = 2\log(\mathrm{hookprod}(\lambda)) + \sum_{k\ge 1} \log(\lambda_k!).$$

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What remains?

To control a sum like

$$\sum_{\lambda \vdash n} e^{-\Psi(\lambda)},$$

it suffices to control

- how small an individual value of $\Psi(\lambda)$ can be, and
- how many partitions $\lambda \vdash n$ there are.

The exact technical statement needed

Everything now reduces to the following deterministic theorem about partitions.

Theorem (Boretsky–Cornejo–H–Horn–Lesnevich–McAllister, *2026*)

For every partition $\lambda \vdash n$,

$$\Psi(\lambda) \geq \frac{3}{2}\, n \log n - 5n.$$

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If one has this bound, then

$$\mathbb{P}(\sigma_1 \preceq_W \sigma_2) \leq n!\, p(n)\, \exp\left(-\frac{3}{2}\, n \log n + 5n\right),$$

where $p(n)$ is the number of partitions of n.

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Using Hardy–Ramanujan,

$$p(n) = \exp(O(\sqrt{n})),$$

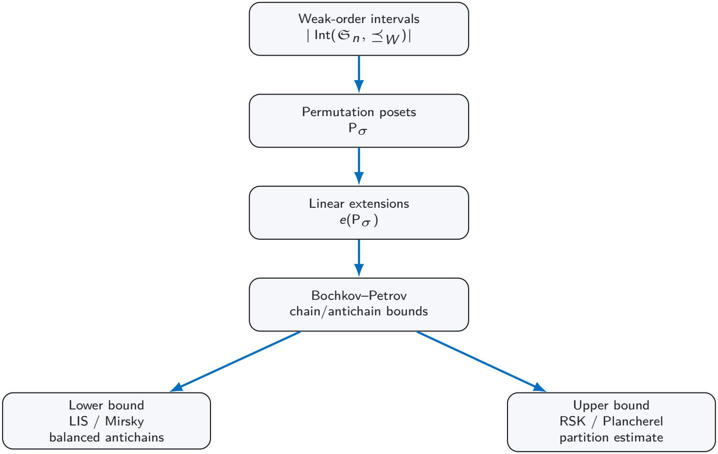
and Stirling,

$$n! = \exp(n \log n - n + O(\log n)),$$

we conclude

$$\mathbb{P}(\sigma_1 \preceq_W \sigma_2) \leq \exp\left(\left(-\tfrac{1}{2} + o(1)\right) n \log n\right).$$

A tour through algebraic combinatorics



Future directions

Strong Bruhat order

Our paper also improves the previous lower bound for strong Bruhat-order comparability. Christo–Michelen have since given the sharp scale: $\mathbb{P}(\sigma_1 \leq_B \sigma_2) = \exp(-\Theta(\log^2 n))$.

Other Dynkin types

Weak order exists for every finite Coxeter group. In other types there are RSK-type correspondences, for example via domino tableaux in type B/C. Can the same Plancherel/shape-averaging program give sharp asymptotics?

Typical intervals and the VKLS window

A random permutation has an RSK shape near the Vershik–Kerov–Logan–Shepp limit curve. But interval counting does not weight all shapes equally. In our upper bound, a shape λ is weighted by

$$\mathbb{P}_{\mathrm{Pl}}(\lambda) \cdot \frac{1}{\prod_k \lambda_k!}.$$

Is the main contribution still coming from typical RSK shapes, or from rare shapes that support unusually large weak-order intervals?

Exact formulas?

Do the asymptotics tell us anything about what an exact formula should look like?

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